Sub:- Basic Civil Engineering:-

* Introduction of civil Engineering:-

→ Civil Engineering is a professional engineering discipline that deals with the design, Construction and maintenance of the physical and naturally built environment, including public works such as road, bridges, canals, dams, airport, railway, sewage, pipeline, Structural Component of building and water Treatment plant.

→ It is Consitered as the second-oldest engineering discipline after military engineering.

→ Civil engineering is traditionally broken into a number of sub disciplines. such as.

i) Construction Engineering
ii) Earthquake Engineering
iii). Environmental Engineering
iv) Forensic Engineering
v) Geotechnical Engineering
vi) Material Science & Engineering
vii). Site developement & planning
viii) Structural Engineering
ix). Surveying
x). Transportation Engineering
xi). Municipal & Urban Engineering
xii). Water resource Engineering
xiii) Civil Engineering System.
xiv) river Engineering & many more

Surveying:-

→ Surveying is the process by which a surveyor measures certain dimensions that occur on or near the surface of the earth.

→ Surveying equipment such as levels and theodolites are used for accurate measurement of angular deviation, horizontal, vertical and slop distances.

→ Data Collected by survey measurement is Converted into a graphical representation of the earth surface in the form of the map, this information is then used by civil Engineers, Contractors and realtors to design, form, build on, and trade respectively.

Transportation Engineering:-

→ Transportation Engineering involves specifying, designing, Constructing and maintaining transportation infrastructure which includes streets, Canals, highways, rail systems, airports, ports and mass transit.

Environmental Engineering:-

→ Environmental Engineering deals with treatment of Chemical, biological or thermal wastes, purification of water and air and remediation of Contaminated sites after waste disposal or accidental Contamination.

* Scope & Outcome of Civil Engineering:-

→ The main Scope of civil Engineering is planning, designing, estimating, supervising, managing and maintenance of structures like building, roads, bridges, dams etc.

* Role of civil Engineers in our Society:-

→ Civil Engineers are Considered to be indispensable professionals. They have one of the gratest responsibilities on this planet, i.e build the quality of life with technical as well as creative skills, these professional plans, design, build and operate the infrastructure necessary for modern civilization, ranging from public water systems and public amenities to highways & bridges.

→ We not may truly sense the significance of civil Engineering in our daily routine. The function of civil Engineering begins with the start of the day when you get down to your routine oral hygiene.

→ The water which you use is delivered through a water supply system with a well designed network of pipes, water treatment plant and many other important things associated with this.

→ The efficient network of road that we use to get around the city is a product of Civil Engineering. Even the energy we need at home and in our business premises is available to us due to the direct Contribution of civil engineers who built the towers for the transmission.

→ A Civil Engineer has the responsibility to design safe structures that meet standard Codes in the regions where they are being built. They must ensure these structures are designed with efficiency, safety, maintenance, and durability in mind.

* Impact of Infrastructural developement on economy of Country:-

→ The lack of infrastructure is hindering the economic growth in many developing Country. Infrastructure Investment has the effects of Contributing to increase the productivity and it is expected to Contribute to future economic growth in developing Contries where infrastructure is still insufficient. Therefore, infrastructure developement is one of the most integral parts of the public policies in developing Contries.

→ Good infrastructure helps to raise productivity and lower cost in the directly productive activities of the economy, but it has to be expanded fast enough to meet the demand for infrastructure in the early stage of developement.

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